

1-merkle_integrated_slicer.py

Front-End User Instructions

Overview

`1-merkle_integrated_slicer.py` is a demonstration slicer and manufacturing accountability prototype.

The program allows a user to:

- Define a simple printable object
- Generate a G-code file
- Create manufacturing records
- Organize generated jobs into structured folders
- Create append-only print logs for auditability

The slicer itself is intentionally simple and primarily serves as a proof-of-concept for the accountability and audit workflow.

Launching the Program

Run the script from a terminal:

```
python 1-merkle_integrated_slicer.py
```

A Tkinter graphical interface window will appear.

Interface Sections

The interface is divided into several sections.

Part + Print Settings

This section contains: - basic object settings - material information - manufacturing identity fields

Most slicer parameters can remain at their defaults for demonstration purposes.

The most important fields are:

Operator ID

Example:

`operator_demo`

This identifies the person operating the printer.

You may use: - a username - initials - employee ID - nickname - workstation name

Example:

`cool_guy`

Facility ID

Example:

`facility_demo`

This identifies the manufacturing location or organization.

Examples:

`ForgeWorks`

`Lab_A`

`Makerspace_01`

`PrototypeDivision`

The Facility ID becomes part of the generated folder structure.

Instrument ID

Example:

`Instrument_demo`

This identifies the specific printer or machine.

Examples:

`Printer01`

`MetalPrinterA`

`MK4_Unit3`

`ForgeBot_07`

This allows multiple printers at the same facility to maintain separate manufacturing histories.

Structured Jobs Root

This section displays the automatically generated output location.

The program builds folders automatically using:

```
output/<facility_id>/<instrument_id>/jobs/
```

Example:

```
output/ForgeWorks/Printer01/jobs/
```

This path updates automatically when the Facility ID or Instrument ID changes.

Optional STL Input

This section allows you to load a simple STL file.

The slicer can: - analyze basic STL dimensions - estimate object type - include STL hash data in the manufacturing records

For demonstration purposes, STL loading is optional.

You may also use the built-in primitive shapes.

Preview + Slice

Press:

```
Preview + Slice
```

to generate a temporary G-code preview.

The generated G-code appears in the preview window on the right side of the interface.

This step does NOT save files yet.

Export G-code + Log

Press:

```
Export G-code + Log
```

to create the manufacturing job.

This performs several actions automatically:

- Generates G-code

- Creates a structured job folder
 - Creates a manifest file
 - Creates or updates the append-only print log
 - Embeds hash information into the G-code
-

Generated Folder Structure

The program automatically creates folders like:

```
output/  
└─ ForgeWorks/  
    └─ Printer01/  
        └─ jobs/  
            └─ 2026-05-24/  
                └─ job_001_cube_20260524_153045/
```

Each job folder contains:

G-code file

Example:

```
cube_20260524_153045.gcode
```

Manifest file

Example:

```
cube_20260524_153045_manifest.json
```

This contains: - job hashes - settings hashes - manufacturing metadata - STL hash information - estimated material usage - estimated print time

Append log

Example:

```
print_log.json
```

This is an append-only chained manufacturing log.

Each new entry references the previous entry hash.

Removing or modifying entries breaks the chain consistency.

Finding Generated Files

Generated files are located in the program's:

`output/`

folder.

The full path is:

`output/<facility_id>/<instrument_id>/jobs/<date>/<job_folder>/`

Example:

`output/ForgeWorks/Printer01/jobs/2026-05-24/job_001_cube_20260524_153045/`

You can open these folders directly in File Explorer.

Recommended Usage

For demonstrations:

1. Choose a Facility ID
2. Choose an Instrument ID
3. Choose an Operator ID
4. Click:

`Preview + Slice`

5. Click:

`Export G-code + Log`

6. Open the generated output folders
-

Purpose of the Demonstration

This slicer is not intended to compete with professional slicers.

Its purpose is to demonstrate:

- manufacturing accountability
- append-only logging
- cryptographic job identity
- structured audit records

- facility/instrument separation
- traceable manufacturing workflows

The generated files can later be: - batched into Merkle trees - publicly anchored - packaged into audit bundles - remotely verified through the audit portal